

Free Resolutions

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This is a note discussing the basic construction of Free Resolutions and the computation of free resolutions for various modules.

1 Basic Construction

Let R be a commutative ring with 1 and let M be a finitely generated module over R . If M is generated by $x_1, \dots, x_n$ elements with m relations f_i between the generators then recall that we can encode the presentation of M with an exact sequence

$$R^m \rightarrow R^n \rightarrow M \rightarrow 0$$

where $R^n \rightarrow M$ is given by $a_i \rightarrow x_i$ and where $R^m \rightarrow R^n$ is given by $b_i \rightarrow f_i(a_i)$. Having a presentation for a module is already pretty good, but we can go further. Recalling that $R\text{-mod}$ is an abelian category gives us that $\ker(R^m \rightarrow R^n)$ is also an R -module. And so we can ask for a presentation of $\ker(R^m \rightarrow R^n)$. This gives us a further exact sequence

$$R^k \rightarrow R^m \rightarrow R^n \rightarrow M \rightarrow 0.$$

We can then iterate this procedure. This process may not terminate (even when M is fg? we should find an example). Repeating this procedure gives the *free resolution* of M

$$\dots \rightarrow R^{k_n} \rightarrow \dots R^{k_2} \rightarrow R^{k_1} \rightarrow R^{k_0} \rightarrow M \rightarrow 0.$$

This is called “Free” since every object in the chain (except the final one) is a free module.

Practically speaking, here’s another way to think about the iteration of this process which may help with the computations. M is finitely generated with n_1 generators and so we start with the exact

sequence

$$R^{n_1} \rightarrow M \rightarrow 0.$$

Now we ask for the generators of $\ker(R^{n_1} \rightarrow M)$; say this has n_2 generators. We can then extend our exact sequence

$$R^{n_2} \rightarrow R^{n_1} \rightarrow M \rightarrow 0$$

Now, ask for the generators of $\ker(R^{n_2} \rightarrow R^{n_1})$,

$$R^{n_3} \rightarrow R^{n_2} \rightarrow R^{n_1} \rightarrow M \rightarrow 0.$$

Then continue this process indefinitely, or until $\ker(R^{n_{k+1}} \rightarrow R^{n_k}) = 0$ is the zero module¹.

Below we will compute some examples of free resolutions of various modules.

2 Examples

These examples are sorted into an order of increasing complexity. However, in all the following examples, R is a polynomial ring and M is a quotient of R . One can check these examples in Macaulay2 with the code, for example,

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kk = ZZ/32003;
R = kk[x,y];
I = ideal(x^2, y^2);
M = R/I;
res M
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Example 1. Let $R = k[x]$ and $M = R/(x^n)$ for some fixed n . Then notice that M is generated by $\bar{1} \in R/(x^n)$ as an R -module. Moreover, we only have a single relation $x^n \cdot \bar{1} = 0$. We can then capture this in the presentation

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

The kernel of $R^1 \rightarrow M$ is exactly the ideal (x^n) . Interpreting (x^n) as an R -module we find that it is a rank 1 free module (generated by x^n). That is a presentation for (x^n) as an R -module is

$$0 \rightarrow R^1 \xrightarrow{1 \mapsto x^n} (x^n).$$

Now since $\ker(R^1 \rightarrow (x^n)) = 0$ we can no longer iterate our procedure. Alternatively $\ker(R^1 \xrightarrow{1 \mapsto x^n} (x^n)) = 0$

¹Note that at each stage of the above construction we are taking the kernels to be finitely generated. This is only necessarily true if R is Noetherian.

$R^1) = 0$. This leaves us with the free resolution of $M = R/(x^n)$.

$$R^1 \xrightarrow{1 \mapsto x^n} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

This example shows us that whenever our kernel module is free then we are done.

Example 2. Let $R = k[x, y]$ and $M = R/(x^2, y^2)$, elements in M are of the form $f = a_0 + a_1x + a_2y + a_3xy$. As an R -module, M is generated by $\bar{1} \in M$. We have the relations $x^2 = 0$ and $y^2 = 0$, in fact, $\ker(R^1 \rightarrow M)$ is exactly the ideal (x^2, y^2) . A presentation for M is thus

$$R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0$$

Now, the kernel module $\ker(R^1 \rightarrow M) = (x^2, y^2)$. This module is generated by two elements x^2, y^2 and we have the R -linear relation that $y^2(x^2) - x^2(y^2) = 0$. Thus²our next iterated sequence is

$$R^1 \xrightarrow{\begin{bmatrix} y^2 \\ -x^2 \end{bmatrix}} R^2 \xrightarrow{\begin{bmatrix} x^2 & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

Now notice that the kernel module $\ker(R^1 \rightarrow R^2) = 0$ is the zero module, and so we are done. The above sequence is thus the free resolution for $M = R/(x^2, y^2)$.

Example 3. Let $R = k[x, y]$ and $M = R/(x, y)^2$. Note $(x, y)^2 = (x^2, xy, y^2)$. Now M is generated as an R -module by $\bar{1} \in M$ but we have the relations $x^2 = 0, xy = 0, y^2 = 0$. This gives the presentation

$$R^3 \xrightarrow{\begin{bmatrix} x^2 & -xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1 \mapsto \bar{1}} M \rightarrow 0.$$

We next want to understand the kernel module $\ker(R^3 \rightarrow R^1)$. If $(a, b, c) \in R^3$ then this triple is in the kernel module when $ax^2 + bxy + cy^2 = 0$. Notice that the two vectors $(y, -x, 0)$ and $(0, y, -x)$ are both vectors³ in the kernel module. We claim that these are in fact an R -basis of $\ker(R^3 \rightarrow R^1)$. The argument we have is a direct computation and is a little tedious. For clarity we will put the details in the appendix and continue the remainder of the argument for computing the free resolution.

We have that $\ker(R^3 \rightarrow R^1)$ is spanned by the vectors $(y, -x, 0), (0, y, -x)$ and so we extend our

²To be rigorous, one needs to argue that for any $(a, b) \in R^2$ such that $ax^2 + by^2 = 0$ that we can write (a, b) as an R -linear combination of (y^2, x^2) . We will do an example of this calculation in the next example.

³These are called “syzigies” and are understood as relations among the relations.

exact sequence

$$R^2 \xrightarrow{\begin{bmatrix} y & 0 \\ -x & y \\ 0 & -x \end{bmatrix}} R^3 \xrightarrow{\begin{bmatrix} x^2 & -xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1-\bar{1}} M \rightarrow 0.$$

Now, continuing the process, we want to understand $\ker(R^2 \rightarrow R^3)$ where $R^2 \rightarrow R^3$ is the map $(a, b) \mapsto (ay, -ax + by, -bx)$. That is, we want to understand for which $[a, b]^T \in R^2$ we have

$$\begin{bmatrix} ay \\ -ax + by \\ -bx \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

However, since R is a domain, we have that this is only satisfied by $a = b = 0$. Thus, $\ker(R^2 \rightarrow R^3) = 0$ and we are done. Our free resolution of $M = R/(x, y)^2$ is thus

$$R^2 \xrightarrow{\begin{bmatrix} y & 0 \\ -x & y \\ 0 & -x \end{bmatrix}} R^3 \xrightarrow{\begin{bmatrix} x^2 & xy & y^2 \end{bmatrix}} R^1 \xrightarrow{1-\bar{1}} M \rightarrow 0.$$

Appendix Calculation. Here we provide the details of the argument that $\ker(R^3 \rightarrow R^1) = \langle (y, -x, 0), (0, y, -x) \rangle$. Suppose $(a, b, c) \in R^3$ such that $ax^2 + bxy + cy^2 = 0$. For this relation to hold true we must have $y|a$, otherwise the term ax^2 could not be cancelled by the other terms. A similar argument gives that we must have $x|c$. Write $a = fy$ and $c = gx$ for $f, g \in R$.

Let $\phi : R^3 \rightarrow R^1$ denote our R -linear map; i.e. $\phi(j, k, l) := jx^2 + kxy + ly^2$. Consider the vector

$$\begin{aligned} \underline{x} &= \begin{bmatrix} a \\ b \\ c \end{bmatrix} - f \begin{bmatrix} y \\ -x \\ 0 \end{bmatrix} - g \begin{bmatrix} 0 \\ y \\ -x \end{bmatrix} \\ &= \begin{bmatrix} fy \\ b \\ gx \end{bmatrix} - f \begin{bmatrix} y \\ -x \\ 0 \end{bmatrix} - g \begin{bmatrix} 0 \\ y \\ -x \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ b + fx - gy \\ 0 \end{bmatrix}. \end{aligned}$$

Now, by R -linearity of ϕ we have that $\phi(\underline{x}) = 0$, in otherwords $xy(b + fx - gy) = 0$. Now, since R is

a domain, we must actually have that $b + fx + gy = 0$ is a relation. Thus $\underline{x}$ is actually the 0 vector in R^3 . That is, given $(a, b, c) \in \ker(R^3 \rightarrow R^1)$ we have shown that there exists $f, g \in R$ such that $(a, b, c) = f(y, -x, 0) + g(0, y, -x)$. That is $(y, -x, 0), (0, y, -x)$ span $\ker(R^3 \rightarrow R^1)$. Moreover, since R is an integral domain, we have that $(y, -x, 0), (0, y, -x)$ are also R -linearly independent, and so in fact, $(y, -x, 0), (0, y, -x)$ is a basis for $\ker(R^3 \rightarrow R^1)$.